



Overview of Respiratory Failure

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Acute respiratory failure is a life-threatening impairment of oxygenation, carbon dioxide elimination, or both.

The respiratory system oxygenates and eliminates carbon dioxide from venous blood. Thus, a useful classification of respiratory failure is:

- [Acute hypoxemic respiratory failure](#), caused by inadequate oxygenation
- [Ventilatory failure](#) (hypercapnic respiratory failure), caused by inadequate carbon dioxide elimination
- [Other causes of respiratory failure](#)

Many disorders affect both oxygenation and ventilation.

Common manifestations include dyspnea, use of accessory muscles of respiration, tachypnea, tachycardia, diaphoresis, cyanosis, altered consciousness, and, without treatment, eventually obtundation, respiratory arrest, and death.

Diagnosis is clinical, supplemented by measurements of arterial or venous blood gases (ABGs or VBGs) and/or peripheral capillary oxygen saturation, and chest radiography.

Although temporizing measures exist, definitive management of respiratory failure frequently necessitates an intensive care unit, and involves correction of the underlying disorder, supplemental oxygen, and ventilatory assistance if needed, possibly with invasive mechanical ventilation, noninvasive [mechanical ventilation](#), or high-flow oxygen.